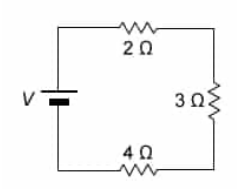


Three resistors are connected in series with a battery as shown.



The 3 Ω resistor is removed from the circuit. How does the voltage across the 4 Ω resistor change?

- ☐ A. The voltage will decrease, because the net resistance of the circuit decreases. [27%]
- ☐ B. The voltage will not change, because the 4 Ω resistor is still in series with the battery. [9%]
- ☐ C. The voltage will not change, because the battery voltage  $V$  is unchanged. [18%]
- ☒ D. The voltage will increase, because the sum of all voltage drops around a closed loop is zero. [44%]

Correct  
43% answered correctly

Explanation:

Applying Kirchhoff's loop rule to determine the voltage drop across a resistor

$V = \Delta V_2 + \Delta V_3 + \Delta V_4$

- $V$  Battery voltage
- $\Delta V_2$  Voltage drop across 2 Ω resistor
- $\Delta V_3$  Voltage drop across 3 Ω resistor
- $\Delta V_4$  Voltage drop across 4 Ω resistor

Initial resistor with current  $I$   
Loop rule and Ohm's law  $\Rightarrow V = 2I + 3I + 4I = 9I$

After removing 3 Ω resistor (new current  $I'$ )  
Loop rule and Ohm's law  $\Rightarrow V = 2I' + 4I' = 6I'$

$\Delta V_2 = \frac{2}{9}V$   
 $\Delta V_3 = \frac{3}{9}V = \frac{1}{3}V$   
 $\Delta V_4 = \frac{4}{9}V$

$\Delta V_2 = \frac{2}{6}V = \frac{1}{3}V$   
 $\Delta V_4 = \frac{4}{6}V = \frac{2}{3}V$

Voltage drop across 4 Ω resistor is greater after removing 3 Ω resistor.

Kirchhoff's loop rule states that the sum of the voltage drops ( $\Delta V_i$ ) around any closed loop in a circuit is zero:

$$\sum_i \Delta V_i = \Delta V_1 + \Delta V_2 + \dots = 0$$

Furthermore, according to Ohm's law, the voltage drop across a resistor is equal to the product of the current  $I$  flowing through the resistor and the resistance  $R$ :

$$\Delta V = IR$$

In this question, the three resistors are initially in series with a battery with voltage  $V$ . Kirchhoff's loop rule implies  $V$  is equal to the sum of voltage drops  $\Delta V_2$ ,  $\Delta V_3$ , and  $\Delta V_4$  across the 2 Ω, 3 Ω, and 4 Ω resistors, respectively:

$$V = \Delta V_2 + \Delta V_3 + \Delta V_4$$

Substituting using Ohm's law gives:

$$\begin{aligned} V &= 2I + 3I + 4I \\ V &= 9I \end{aligned}$$

Hence, the ratio of  $\Delta V_4$  to the battery's  $V$  is equal to:

$$\frac{\Delta V_4}{V} = \frac{4I}{9I}$$

Solving for  $\Delta V_4$  gives

$$\Delta V_4 = \frac{4}{9}V$$

When the 3 Ω resistor is subsequently removed, the loop rule equation and Ohm's law imply that:

$$V = 2I' + 4I' = 6I'$$

where  $I'$  is the current in the new circuit. With fewer resistors in the loop, each resistor will experience a larger fraction of the voltage of the battery. The ratio of the voltage drop across the 4 Ω resistor ( $\Delta V_4'$ ) to the battery's  $V$  is:

$$\frac{\Delta V_4'}{V} = \frac{4I'}{6I'}$$

Solving for  $\Delta V_4'$  yields:

$$\Delta V_4' = \frac{4}{6}V = \frac{2}{3}V > \frac{4}{9}V$$

Therefore, the loop rule implies that the voltage drop across the 4 Ω resistor **increases** after removing the 3 Ω resistor.

(Choice A) The total resistance of the circuit decreases. However, decreasing resistance increases current and voltage across the remaining resistors.

(Choice B) The 4 Ω resistor is still in series, but because there are fewer resistors, each remaining resistor must have a larger voltage drop.

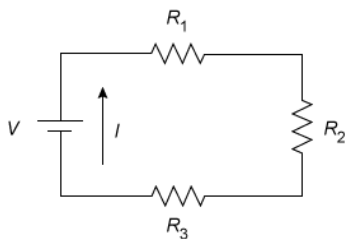
(Choice C) The battery voltage is unchanged, but removing a resistor from the circuit will change the voltages across the remaining resistors.

Educational objective:

The sum of voltage drops around a closed loop is zero. When a resistor is removed from a series circuit, there is a larger voltage drop across the remaining resistors.

**Time spent:**0 sec  
**Subject:**  
Physics

**QID:**404013  
**System:**  
Electrostatics and Circuits



Three resistors are connected in series with a battery with voltage  $V$ . If  $R_1 > R_2 > R_3$ , which of the following expressions correctly shows the relationship between the battery voltage and the voltages across the resistors?

- ✓ ☒ A.  $V = IR_1 + IR_2 + IR_3$  [75%]  
☐ B.  $IR_1 > IR_2 > IR_3 > V$  [8%]  
☐ C.  $IR_3 < IR_1 < IR_2 < V$  [5%]  
☐ D.  $V = IR_1 = IR_2 = IR_3$  [9%]

Correct

76% answered correctly

### Explanation:

Relationship between battery voltage and potential drops across resistors via Kirchhoff's loop rule

Ohm's law  
 $V = IR$

Kirchhoff's loop rule  
 $\sum \Delta V_i = V - IR_1 - IR_2 - IR_3 = 0$

•  $V$  Voltage  
 •  $I$  Current  
 •  $R$  Resistance

$V = IR_1 + IR_2 + IR_3$

$V$  across the battery is positive and  $V_1$  and  $V_2$  and  $V_3$  across the resistors are negative.

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In an electric circuit, a source of voltage generates a **current**  $I$ , which is the flow of charge per unit time in a conductor (ie, wire, bulb, resistor). When circuit elements are in series, they are connected end-to-end in a closed loop. **Conservation of charge** applies to a series circuit, implying the magnitude of current flow is the same through each element.

When the current flows through a conductor (eg, resistor), the voltage will drop across the conductor due to resistance. In any closed loop in an electrical circuit, energy is conserved and the algebraic sum of all voltage drops  $\Delta V_i$  is zero:

$$\sum \Delta V_i = \Delta V_1 + \Delta V_2 + \dots = 0$$

Moreover, **Ohm's Law** states that voltage is equal to the product of resistance  $R$  and current:

$$\Delta V = IR$$

In this question, the **current is the same** through all resistors because they are in series. Also, the voltage across the battery  $V$  is positive and potential differences across resistors are negative. Therefore, applying Kirchhoff's loop rule and Ohm's law:

$$\Delta V_i = V - IR_1 - IR_2 - IR_3 = 0$$

Therefore, the potential difference across the battery is related to the potential differences across the resistors as:

$$V = IR_1 + IR_2 + IR_3$$

**(Choice B)** The voltage across the battery is not less than the voltage across the resistors, and it is equal to the sum of the voltage drop across each resistor.

**(Choice C)**  $R_1$  is larger than  $R_2$ . Therefore,  $IR_1$  is greater than  $IR_2$ .

**(Choice D)** All three resistance values are not equal, so the voltage across each resistor will not be equal, and the battery will have the largest voltage.

### Educational objective:

The sum of all voltages in any closed loop within a circuit is zero. Voltage is the product of resistance and current.